

AC3303 & AC3304

Introduction to Asset Pricing & Valuation

Spring 2023 Examination

Paper 1 · 1.5 hours · Answer Question 1 and one other question · Total 100 Marks

Module	AC3303 & AC3304 — Introduction to Asset Pricing & Valuation
Session	Spring 2023
Paper	1 of 1
External Examiner	Professor Thomas O'Connor
Internal Examiner	Dr. Joe Healy
Structure	Question 1 (Compulsory, 50 Marks) + one of Question 2 or Question 3 (50 Marks)
This scheme covers	Question 1, Question 2 and Question 3 in full, so you can study either route through the paper

Own-figure rule. Where a candidate makes an error early in a multi-step question, full method marks are still awarded for correctly applying the required technique to their own, incorrect figure in every subsequent step. This applies throughout every question in this scheme and is restated wherever it is especially relevant.

How this scheme was built. Every calculation below was computed independently in Python before being written up and cross-checked against internal consistency tests (probabilities summing to 1, portfolio variance built directly from the variance-covariance formula, bond prices checked for the expected direction of change with yield). Discursive answers draw on standard corporate finance theory rather than any one source, with own-figure and alternative-argument credit noted throughout.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

Scenario

Advising a client on shares in **Tours PLC** (tourism sector) and **Logiquick PLC** (medical logistics sector), both listed on the Irish Stock Exchange. Four economic scenarios are given:

State of the Economy	Probability	Return: Tours PLC	Return: Logiquick PLC
Excellent Growth	0.22	10%	8%
Low Level Growth	0.35	4%	6%
No Growth at all	0.23	0%	2%
Negative Growth	0.20	-4%	-1%
Total	1.00		

(a) Required

10 Marks

Calculate the Expected Return and Standard Deviation risk for both shares.

FULL MODEL ANSWER

Expected return is the probability-weighted average return; standard deviation is the square root of the probability-weighted sum of squared deviations from that mean.

State	Prob.	Return Tours	Prob × Return	Return Logiquick	Prob × Return
Excellent Growth	0.22	10%	2.20	8%	1.76
Low Level Growth	0.35	4%	1.40	6%	2.10
No Growth at all	0.23	0%	0.00	2%	0.46
Negative Growth	0.20	-4%	-0.80	-1%	-0.20
E(R)			2.80%		4.12%

$$E(R_{\text{Tours}}) = 2.80\% \quad E(R_{\text{Logiquick}}) = 4.12\%$$

State	Prob.	(R-E(R)) Tours	Sq. × Prob	(R-E(R)) Logiquick	Sq. × Prob
Excellent Growth	0.22	7.20	11.4048	3.88	3.3120
Low Level Growth	0.35	1.20	0.5040	1.88	1.2370
No Growth at all	0.23	-2.80	1.8032	-2.12	1.0337
Negative Growth	0.20	-6.80	9.2480	-5.12	5.2429
Variance			22.96		10.83

Var(Tours) = 22.96 → SD(Tours) = 4.79%

Var(Logiquick) = 10.83 → SD(Logiquick) = 3.29%

Probabilities sum to exactly 1.00 and each squared-deviation column was recomputed from the raw state returns, so no rounding drift enters the variance figures.

Mark Allocation — Part (a), 10 Marks

Tier	Marks	What separates this tier
9-10	9-10	Both expected returns and both standard deviations fully correct with clear probability-weighted working shown for each share.
6-8	6-8	Both expected returns correct; one standard deviation correct and the other has a minor arithmetic slip carried through consistently (own-figure).
3-5	3-5	Correct method applied but with a computational error affecting more than one figure.
0-2	0-2	Wrong technique used or no workable attempt.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(b) Required**5 Marks**

Calculate the co-variance between returns on shares in Tours Plc and on shares in Logiquick Plc.

FULL MODEL ANSWER

State	Prob.	Dev. Tours	Dev. Logiquick	Product	Prob × Product
Excellent Growth	0.22	7.20	3.88	27.936	6.1459
Low Level Growth	0.35	1.20	1.88	2.256	0.7896
No Growth at all	0.23	-2.80	-2.12	5.936	1.3653
Negative Growth	0.20	-6.80	-5.12	34.816	6.9632
Covariance					15.264

Cov(Tours, Logiquick) = 15.264

The positive covariance means the two shares' returns move together across economic states, consistent with both being sensitive to the same broad economic cycle even though they operate in different sectors.

Mark Allocation — Part (b), 5 Marks

Tier	Marks	What separates this tier
5	5	Correct covariance with full working shown.
3-4	3-4	Correct method but a minor own-figure carry-through error from part (a).
1-2	1-2	Some correct elements but incorrect combination or formula.
0	0	No credible attempt.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(c) Required

6 Marks

Calculate the correlation coefficient between returns on shares in Tours Plc and on shares in Logiquick Plc. Explain what this correlation coefficient means.

FULL MODEL ANSWER

$$\rho = \text{Cov} \div (\text{SD}_{\text{Tours}} \times \text{SD}_{\text{Logiquick}}) = 15.264 \div (4.79 \times 3.29) = 0.968$$

Strength and direction: A correlation of 0.968 is very close to the maximum of +1, indicating an almost perfectly positive linear relationship between the two shares' returns despite them operating in different sectors.

Diversification implication: Because the two shares move almost in lockstep, combining them provides very little diversification benefit — portfolio risk will sit close to the weighted average of the two individual standard deviations rather than being meaningfully reduced.

Economic interpretation: The high correlation suggests both companies' returns are driven predominantly by the same broad macroeconomic cycle (the four economic states given) rather than by sector-specific factors independent of one another.

Mark Allocation — Part (c), 6 Marks

Tier	Marks	What separates this tier
6	6	Correct correlation coefficient (own-figure) plus explanation covering both strength/direction and the diversification implication.
4-5	4-5	Correct coefficient with a partial explanation.
2-3	2-3	Correct method but computational error, or correct coefficient with no meaningful explanation.
0-1	0-1	Incorrect formula or no attempt.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(d) Required**7 Marks**

Determine the Expected Return and Standard Deviation risk of a portfolio where the investor decides to invest one quarter of their money in Tours Plc and the remainder in Logiquick Plc.

FULL MODEL ANSWER

One quarter in Tours means $w_{\text{Tours}} = 25\%$, $w_{\text{Logiquick}} = 75\%$.

Component	Value
Weight in Tours (w_1)	0.25
Weight in Logiquick (w_2)	0.75
$E(R_p) = 0.25(2.80\%) + 0.75(4.12\%)$	3.79%
$w_1^2\sigma_1^2 = 0.25^2 \times 22.96$	1.4350
$w_2^2\sigma_2^2 = 0.75^2 \times 10.83$	6.0894
$2w_1w_2\text{Cov} = 2 \times 0.25 \times 0.75 \times 15.264$	5.7240
Portfolio Variance σ_p^2	13.2484
Portfolio Standard Deviation σ_p	3.64%

With correlation this high (0.968), the portfolio standard deviation (3.64%) sits between the two individual standard deviations (4.79% and 3.29%) rather than meaningfully below the lower of the two — the expected signature of a near-perfectly correlated pairing, confirming the covariance term has been applied correctly.

Mark Allocation — Part (d), 7 Marks

Tier	Marks	What separates this tier
6-7	6-7	Correct portfolio expected return and correct portfolio standard deviation using the full variance-covariance formula.
4-5	4-5	Correct portfolio return; variance formula correct in structure but with an arithmetic slip or misapplied covariance term (own-figure applied thereafter).
2-3	2-3	Portfolio return correct only, or variance calculated as a simple weighted average with no covariance term.
0-1	0-1	No credible attempt.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(e) Required**22 Marks**

In the context of the Efficient Markets Hypothesis, explain what is meant by 'Stock Market Efficiency'. In addition, explain the value of an efficient market in terms of encouraging participation in Capital Markets and the signals that an efficient market gives to management.

FULL MODEL ANSWER

Stock market efficiency (the Efficient Market Hypothesis, EMH) holds that security prices fully and rapidly reflect all available information, so no investor can consistently earn abnormal (risk-adjusted) returns using that information. Three forms are distinguished by the information set assumed to be reflected in price:

Form	Information set reflected in price	Implication
Weak form	All past price and volume data	Technical analysis cannot generate abnormal returns
Semi-strong form	All publicly available information	Fundamental analysis of public information cannot generate abnormal returns
Strong form	All information, including private/insider information	Even insiders cannot consistently outperform the market

Question 1 (continued)

Value of an efficient market in encouraging participation in Capital Markets

Fair pricing builds investor confidence: If investors believe prices are fair and not systematically manipulable by better-informed insiders, they are more willing to commit savings to capital markets rather than holding cash or less productive assets, which widens the pool of capital available to fund companies.

Lower cost of capital economy-wide: Greater participation and liquidity resulting from investor confidence reduces the return investors demand for bearing risk (the liquidity and information-risk premium), which in turn lowers the cost of capital for companies raising funds, supporting more investment in the real economy.

A level playing field: Because semi-strong efficiency implies prices reflect public information regardless of an investor's size or sophistication, smaller/retail investors can participate without being structurally disadvantaged relative to large institutions trading on the same public information set.

Signals an efficient market gives to management

The share price is a credible performance scorecard: Because the market price rapidly incorporates public information about the company, management can treat share price reactions to their decisions (e.g. an investment announcement) as a genuine, largely unbiased signal of whether the market believes that decision creates value.

No reward for cosmetic accounting: Since the market can see through purely cosmetic reporting choices that don't change underlying cash flows, management gains little from earnings management — a signal that resources are better spent on substantive value creation than on managing short-term reported numbers.

Financing and payout decisions carry information content: Because prices are highly sensitive to new information, actions such as issuing new equity or changing the dividend are themselves interpreted by the market as signals about management's view of the company's prospects — efficient markets are precisely what makes such signalling effects strong and fast-acting.

Other acceptable answers. Candidates who define EMH and its three forms accurately, and develop distinct points under both the "encouraging participation" and "signals to management" headings with sound economic reasoning, should receive full credit regardless of the specific points chosen or their ordering.

Mark Allocation — Part (e), 22 Marks

Tier	Marks	What separates this tier
18-22	18-22	Clear, accurate definition of EMH and its forms; well-developed discussion of BOTH the value in encouraging market participation AND the signals given to management.
13-17	13-17	Good definition; both required strands (participation and management signals) addressed but one noticeably thinner than the other.
8-12	8-12	Definition present but incomplete or imprecise; discussion largely one-sided (only participation, or only management signals).
0-7	0-7	Definition of EMH absent or materially incorrect; no meaningful discussion of either required strand.

Question 1 Total: 50 Marks

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

How to use this scheme. The paper requires Question 1 plus **one** of Question 2 or Question 3 — both are marked out of 50 and count equally toward the paper. This scheme covers both in full so you can revise whichever route you are more comfortable with.

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Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(a) Required**8 Marks**

Calculate the numbers a)–e) to complete the following table (Present Value, Years, Interest Rate, Future Value):

FULL MODEL ANSWER

Every row uses $FV = PV(1+r)^n$, rearranged for whichever variable is missing.

Present Value	Years	Interest Rate	Future Value
500	1	3%	a) ?
500	10	7%	b) ?
c) ?	4	2%	110,000
9,000	9	d) ?	18,000
30,000	10	e) ?	90,000

Row	Formula used	Answer
a)	$FV = 500 \times (1.03)^1$	515.00
b)	$FV = 500 \times (1.07)^{10}$	983.58
c)	$PV = 110,000 \div (1.02)^4$	101,623.00
d)	$r = (18,000/9,000)^{1/9} - 1$	8.01%
e)	$r = (90,000/30,000)^{1/10} - 1$	11.61%

Each answer was independently verified in Python using the same $FV = PV(1+r)^n$ relationship rearranged algebraically.

Mark Allocation — Part (a), 8 Marks

Tier	Marks	What separates this tier
7-8	7-8	All five values (a–e) correct, including both rate-solving rows.
5-6	5-6	Three or four values correct.
2-4	2-4	One or two values correct; correct general formula recognised but frequent rearrangement errors.
0-1	0-1	Formula not recognised or applied incorrectly throughout.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(b) Required

12 Marks

An 8-year U.S. Treasury bond with a face value of \$1,000 pays a coupon of 6.50% paid semi-annually. The reported yield to maturity is 3.0%.

- What is the present value of the bond? (4 Marks)
- If the yield to maturity changes to 4%, what would be the present value? (4 Marks)
- If the yield to maturity changes to 2%, what would be the present value? (4 Marks)

FULL MODEL ANSWER

Because the coupon is paid **semi-annually**, both the coupon and the discount rate must be halved, and the number of periods doubled: $n = 16$ half-years, semi-annual coupon = $6.50\%/2 \times \$1,000 = \32.50 , semi-annual rate = $YTM/2$.

$$\text{Price} = C \times [1 - (1+r)^{-n}] / r + FV / (1+r)^n$$

Yield to Maturity	PV of Coupons	PV of Face Value (\$1,000)	Bond Price
3.0% (given)	459.27	788.03	\$1,247.30
4.0%	441.28	728.45	\$1,169.72
2.0%	478.33	852.82	\$1,331.15

Sense check. The coupon rate (6.50%) exceeds every YTM tested, so the bond should trade at a premium to face value throughout — all three prices are comfortably above \$1,000. The inverse YTM/price relationship holds throughout: price falls as YTM rises from 2% → 3% → 4%.

Mark Allocation — Part (b), 12 Marks (4 per YTM scenario)

Tier	Marks	What separates this tier
4 (per scenario)	4	Correct bond price with semi-annual coupon and rate correctly applied, coupon annuity PV and face value PV shown separately and summed.
2-3 (per scenario)	2-3	Correct formula structure but an arithmetic error, or annual compounding used instead of semi-annual (own-figure).
1 (per scenario)	1	One component (coupon annuity or face value PV) correct in isolation but not combined.
0 (per scenario)	0	Wrong formula or no attempt.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(c) Required

20 Marks

It has been argued that The Capital Asset Pricing Model has certain limitations in terms of its usefulness as well as a number of associated problems. Discuss these limitations and problems, referencing any empirical research you may be aware of in this area.

FULL MODEL ANSWER

The CAPM is the standard tool for estimating a required/expected return given systematic risk, but its practical application faces several well-documented problems:

Estimating beta is backward-looking and unstable: Beta is typically estimated from a historical regression of a share's returns against a market index, but a company's risk profile changes over time, so a historical beta may poorly estimate the beta that will actually prevail going forward.

Choice of market proxy (Roll's critique): CAPM requires the return on the true market portfolio — every risky asset in the economy — but in practice a proxy such as a stock index is used, which excludes bonds, property, human capital and unlisted assets. Roll (1977) argued this makes CAPM effectively untestable, since the true market portfolio is unobservable.

Empirical evidence of missing risk factors: Fama and French's empirical work (1992/1993) finds that firm size and the book-to-market ratio help explain the cross-section of stock returns beyond market beta alone, suggesting CAPM's single-factor structure does not fully capture priced risk — leading to multi-factor alternatives.

Risk-free rate and equity risk premium are themselves estimates: Both inputs require judgement — the market risk premium in particular varies substantially depending on the historical period and method used to estimate it, feeding directly into the required-return output.

Restrictive assumptions: CAPM assumes investors can borrow and lend freely at the risk-free rate, face no taxes or transaction costs, and hold well-diversified portfolios with homogeneous expectations — assumptions that do not hold precisely in practice.

Practical impact on applicability: Despite these problems, CAPM remains widely used in practice for cost-of-equity estimation because it is simple, transparent, and its inputs are observable — but the output should be treated as an estimate with a margin of error, ideally cross-checked against alternative approaches (e.g. dividend growth model, multi-factor models).

Candidates may instead or additionally discuss the single-period nature of the model or cite other named empirical studies. Any well-explained subset of the standard critique, correctly linked to practical applicability, should receive full credit.

Mark Allocation — Part (c), 20 Marks

Tier	Marks	What separates this tier
16-20	16-20	At least four to five distinct, well-explained problems, each clearly linked to practical impact on applicability, with reference to empirical research.
11-15	11-15	Three to four relevant problems identified with reasonable explanation, limited or no reference to empirical research.
6-10	6-10	One to two problems identified, or points listed without explanation of why they matter in practice.
0-5	0-5	No accurate discussion of CAPM's limitations.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(d) Required

10 Marks

"A significant amount of effort has been made to address the problems identified by Agency theory in terms of ensuring managers operate in the best interests of stakeholders" Discuss this statement, describing the problems identified by Agency theory and how these can be addressed.

FULL MODEL ANSWER

Agency theory describes the conflict of interest that arises when managers (agents) do not perfectly act in the interests of shareholders and other stakeholders (principals), because their incentives are not automatically aligned.

The core agency problem: Managers may pursue objectives that benefit themselves — empire-building, excessive perks, entrenchment, or short-term actions that boost their own compensation — rather than maximising shareholder wealth, especially where ownership is dispersed and monitoring is costly.

Agency costs: These arise in three forms: monitoring costs (e.g. audits, board oversight) borne by shareholders; bonding costs borne by managers to credibly commit to acting in shareholders' interests (e.g. contractual restrictions); and the residual loss that remains even after monitoring and bonding, representing the value lost from imperfect alignment.

Alignment through incentive design: Performance-related pay, share options, and equity-based compensation tie management's own wealth to the share price, reducing the incentive to act against shareholders' interests.

Governance mechanisms: An independent board of directors, audit committees, and non-executive directors provide oversight and monitoring of management decisions on behalf of shareholders.

Market-based discipline: The threat of takeover (poorly performing companies with undervalued shares become acquisition targets, typically leading to management being replaced) and the market for corporate control more broadly act as an external discipline on management.

Debt as a disciplining mechanism: Higher leverage commits the company to fixed interest payments, reducing the free cash flow available for managers to spend on non-value-maximising projects (the free cash flow hypothesis).

Candidates may instead discuss disclosure/transparency requirements, say-on-pay shareholder votes, or specific governance codes. Any well-explained combination of problem description and alignment mechanisms should receive full credit.

Mark Allocation — Part (d), 10 Marks

Tier	Marks	What separates this tier
8-10	8-10	Clear description of the agency problem and agency costs, with at least three distinct, well-explained mechanisms for addressing it.
5-7	5-7	Agency problem described adequately but fewer than three addressing mechanisms, or mechanisms listed without explanation.
2-4	2-4	Agency problem only vaguely described, with minimal discussion of solutions.
0-1	0-1	No meaningful discussion of agency theory.

Question 2 Total: 50 Marks

Question 3

50 Marks — Choice question: alternative to Question 2

(a) Required

8 Marks

Using the Capital Asset Pricing Model (CAPM) calculate the expected return for Share A, assuming a beta factor of 1.25, a risk free rate of interest of 4%, and a market risk premium of 7%. Explain what is meant by a beta factor.

FULL MODEL ANSWER

$$E(R) = R_f + \beta(\text{Market Risk Premium})$$

Component	Value
Risk-free rate, R_f	4.00%
Beta, β	1.25
Market risk premium	7.00%
$\beta \times \text{Market risk premium} = 1.25 \times 7\%$	8.75%
$E(R) = 4\% + 8.75\%$	12.75%

What a beta factor means: Beta measures a share's sensitivity to movements in the overall market — the systematic (non-diversifiable) risk that cannot be removed by holding a diversified portfolio. A beta of 1.25 means the share's returns tend to move 1.25 times as much as the market's, in the same direction, so Share A is more volatile than the market and carries above-average systematic risk, justifying a required return above the market return of 11% (4% + 7%).

Mark Allocation — Part (a), 8 Marks

Tier	Marks	What separates this tier
7-8	7-8	Correct CAPM calculation and a clear, accurate explanation of beta as a measure of systematic/non-diversifiable risk relative to the market.
4-6	4-6	Correct calculation but explanation of beta vague or incomplete (e.g. no mention of systematic risk / diversification).
1-3	1-3	Formula misapplied, or beta explanation given with no calculation.
0	0	No credible attempt.

Question 3

50 Marks — Choice question: alternative to Question 2

(b) Required

12 Marks

Consider the three stocks outlined below and calculate the stock price for each of the three stocks if the market capitalization rate for each stock is 9.00%.

- Stock A is expected to provide a dividend of \$12.00 a share forever. (3 Marks)
- Stock B is expected to pay a dividend of \$6.00 next year. Thereafter, dividend growth is expected to be 4% a year forever. (4 Marks)
- Stock C is expected to pay a dividend of \$7.00 next year. Thereafter, dividend growth is expected to be 10% a year for four years (i.e., years 2 through 5 inclusive) and zero thereafter. (5 Marks)

FULL MODEL ANSWER

Stock A — level (no-growth) perpetuity

$$\text{Price} = \text{Dividend} \div r = \$12.00 \div 0.09 = \$133.33$$

Stock B — Gordon (constant) growth model

$$\text{Price} = D_1 \div (r - g) = \$6.00 \div (0.09 - 0.04) = \$120.00$$

Stock C — supernormal growth (10% in years 2-5), zero growth thereafter

$D_1 = \$7.00$, then 10% growth applies for years 2 through 5 inclusive, after which the dividend stays flat forever:

Year	Dividend
1	\$7.0000
2 ($\times 1.10$)	\$7.7000
3 ($\times 1.10$)	\$8.4700
4 ($\times 1.10$)	\$9.3170
5 ($\times 1.10$)	\$10.2487

From year 6 onward the dividend is flat at \$10.2487 forever (zero growth), so its value at the end of year 5 is a level perpetuity: $TV_5 = D_5 \div r = 10.2487 \div 0.09 = \113.87

Component	Present Value
PV(D_1) through PV(D_5), discounted at 9%	\$32.70
PV of terminal value ($113.87 \div 1.09^5$)	\$74.01
Stock C Price	\$106.72

Each stock uses a different valuation model matched to its dividend pattern: level perpetuity (A), Gordon growth (B), and a finite growth phase folding into a level perpetuity from year 6 (C). All three were computed independently in Python.

Mark Allocation — Part (b), 12 Marks (3 + 4 + 5)

Tier	Marks	What separates this tier
Full marks	3/4/5	Correct model selected and correctly applied for each stock, with the correct final price.
Partial	varies	Correct model recognised but a computational error (e.g. wrong number of discounting periods for the terminal value in Stock C), own-figure applied thereafter.
Low	1	Dividend correctly identified but wrong valuation model applied (e.g. Gordon growth used for the no-growth perpetuity).
0	0	No credible attempt for that stock.

Question 3

50 Marks — Choice question: alternative to Question 2

(c) Scenario

20 Marks

In March 2023 a U.S. investor holds a well-diversified stock portfolio with a market value of \$10 million. 75% of the portfolio has been constructed to match the performance of the S&P 500 index. The other 25% is held in twenty international shares favoured by the investor. The investor is worried about a significant drop in equity prices over the following six-month period and is considering the options market to hedge.

On March 1 2023 the S&P 500 index stands at 3900. Available S&P 500 index options (all with a \$500 multiplier): June 2023 expiration, strike 3750; September 2023 expiration, strike 3900; December 2023 expiration, strike 3800.

- (i) Recommend a suitable hedging strategy. Can the investor hedge 100% of the portfolio using these options? (4 Marks)
- (ii) Calculate the payoff to such a strategy if the S&P 500 declines by 10% over the next six months. (4 Marks)
- (iii) Outline the factors affecting option prices per the Black-Scholes formula and their effect on call and put prices. (12 Marks)

FULL MODEL ANSWER

(i) Recommended strategy: Buy S&P 500 index **put options** (a protective put on the index-tracking portion of the portfolio) — this locks in a floor value for that portion while preserving upside if the market rises. The **September 2023 expiration** is the correct choice, since March 1 to September is exactly the six-month horizon the investor is worried about (June is too short at ~3 months, December too long at ~9 months). Within the September expiration, the **3900 strike** is at-the-money (equal to the current index level), giving the most direct hedge against the specific "significant drop" the investor fears.

Can 100% of the portfolio be hedged? No. Only the 75% of the portfolio that is constructed to match the S&P 500 can be effectively hedged using S&P 500 index options, because that portion's returns move closely with the index. The remaining 25% is held in twenty individual international shares, which carry idiosyncratic, country-specific and currency risk not captured by the S&P 500 — a US index option cannot hedge risk that isn't correlated with the US index. So the investor can hedge the systematic risk of the 75% index-tracking portion, but not the 25% single-stock portion.

Question 3 (continued)

(ii) Payoff if the S&P 500 falls 10%

Step	Calculation	Result
Value of index-tracking portion	$75\% \times \$10,000,000$	\$7,500,000
Contracts needed	$\$7,500,000 \div (3900 \times \$500)$	3.85 → round to 4 contracts
Index level after 10% fall	3900×0.90	3,510
Put intrinsic value (points)	$\text{Max}(3900 - 3,510, 0)$	390 points
Payoff per contract	$390 \times \$500$	\$195,000
Total payoff (4 contracts)	$4 \times \\$195,000$	\$780,000

Reconciliation. The 75% index-tracking portion loses $10\% \times \$7,500,000 = \$750,000$ in value. Using the exact (unrounded) hedge ratio of 3.846 contracts, the put payoff is exactly \$750,000 — a perfect offset, as expected for an at-the-money put sized precisely to the exposure. Rounding up to a whole number of contracts (4) slightly over-hedges, producing a payoff of \$780,000 against the \$750,000 loss. The 25% international-share portion is unhedged; its gain or loss over the same period is not quantifiable from the data given, since no return is specified for those shares — this is the direct consequence of the "no, not 100%" answer in part (i). No option premium is given in the question, so this payoff is the option's intrinsic value at expiry; the actual net profit on the hedge would be this payoff less whatever premium was paid to buy the puts.

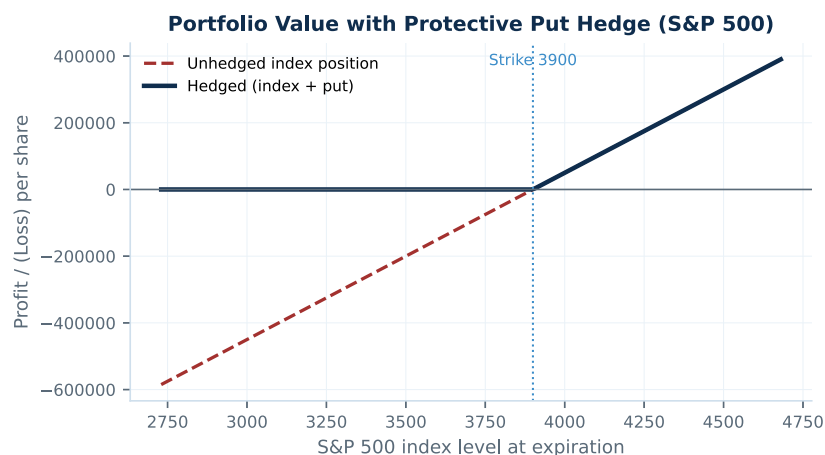


Fig. 1 — Value of the hedged vs unhedged index-tracking position around the September 3900 put

Mark Allocation — Part (c)(i)-(ii), 8 Marks

Tier	Marks	What separates this tier
7-8	7-8	Protective put correctly identified with the correct expiration (September) and strike (3900) reasoned from the data; correctly concludes 100% cannot be hedged with sound reasoning; payoff calculation correct including contract sizing.
5-6	5-6	Correct strategy and correct 'no' on 100% hedging, but expiration/strike choice not clearly reasoned, or a minor error in the payoff calculation.
2-4	2-4	Correct general strategy (buying puts) but wrong expiration/strike, or contract sizing omitted/incorrect.
0-1	0-1	Wrong strategy (e.g. selling options, or claiming 100% can be hedged) or no credible attempt.

Question 3

50 Marks — Choice question: alternative to Question 2

(c)(iii) Required

12 Marks

Outline the factors affecting option prices as specified in the Black-Scholes option pricing formula and briefly discuss the effect of a change in each of these factors on the prices of both call and put options.

FULL MODEL ANSWER

Variable	Effect on Call price	Effect on Put price
Current underlying price (S)	Increases as S rises	Decreases as S rises
Exercise/strike price (K)	Decreases as K rises	Increases as K rises
Time to expiration (T)	Increases with more time	Generally increases with more time
Volatility of the underlying (σ)	Increases with volatility	Increases with volatility
Risk-free interest rate (r)	Increases as r rises	Decreases as r rises

Underlying price and strike price: A call gains value the further the underlying price is above the strike (more intrinsic value); a put gains value the further the underlying price is below the strike — the direct in-the-money effect for each type.

Time to expiration: More time gives more opportunity for a favourable price move before expiry, so both calls and puts are generally worth more with more time remaining (time value), all else equal.

Volatility: Because option payoffs are asymmetric — losses capped at the premium paid, upside uncapped for the holder — higher volatility raises the probability of a large favourable move without symmetrically raising the cost of an unfavourable one, so both call and put values rise with volatility.

Risk-free interest rate: A higher rate raises the cost of carrying the underlying and lowers the present value of the exercise price paid in future, which increases call values and decreases put values.

Candidates who correctly identify all five variables and their qualitative direction of effect on both call and put prices should receive full credit — the numerical Greeks are not required at this level, only the direction of each relationship.

Mark Allocation — Part (c)(iii), 12 Marks

Tier	Marks	What separates this tier
10-12	10-12	All five variables correctly identified with accurate direction of effect on BOTH calls and puts, with sound underlying reasoning.
7-9	7-9	Four or five variables identified with mostly correct direction, minor errors on one variable.
4-6	4-6	Two to three variables correctly identified, or direction of effect frequently confused between calls and puts.
0-3	0-3	Fewer than two variables correctly identified.

Question 3

50 Marks — Choice question: alternative to Question 2

(d) Required

10 Marks

Discuss the main stakeholders in an organisation.

FULL MODEL ANSWER

Stakeholders are any group with an interest in, or affected by, an organisation's decisions and performance:

Stakeholder	Primary interest
Shareholders / owners	Return on investment: dividends and capital growth; ultimate residual claimants on profit
Management / employees	Remuneration, job security, career progression; day-to-day decision-making authority
Debt holders / creditors	Timely payment of interest and repayment of principal; protection of the value of their claim
Customers	Quality, price and reliability of the goods or services provided
Suppliers	Timely payment and a stable, ongoing trading relationship
Government / regulators	Tax revenue, compliance with law and regulation, wider economic and social impact
Local community / society	Employment, environmental impact, and the broader social footprint of the organisation

Why the distinction matters: Shareholder wealth maximisation is the traditional objective of corporate finance, but management must balance shareholders' interests against those of other stakeholders — sustained conflict with any one group can ultimately undermine shareholder value itself (e.g. supplier disputes disrupting operations, or regulatory penalties from ignoring compliance obligations).

Link to agency theory: The stakeholder framework overlaps with agency theory — management (the agent) may not always act purely in shareholders' (the principal's) interest, and shareholders' interests may not always align with other stakeholder groups, which is why governance mechanisms exist to balance these competing claims.

Mark Allocation — Part (d), 10 Marks

Tier	Marks	What separates this tier
8-10	8-10	At least five distinct stakeholder groups identified with their primary interest correctly described, plus discussion of why balancing these interests matters.
5-7	5-7	Three to four stakeholder groups correctly identified and described, but no discussion of why the distinction matters.
2-4	2-4	One to two stakeholder groups identified, or groups listed without describing their interest.
0-1	0-1	No meaningful list of stakeholders.

Question 3 Total: 50 Marks